

Software Design Specification (SDS)

Project Name: SmartStream - Integrated Business Management System

Client: KTD LANKA (PVT) LTD

Group No: 07 - NextGen Developers

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1. Introduction

The Software Design Specification (SDS) details the architectural decisions, structural patterns, object-oriented mapping, and user interface designs required to build the SmartStream system. Transitioning from the requirements outlined in the Software Requirements Specification (SRS), this document serves as the comprehensive technical blueprint for the development team. It defines how the system will meet the client's needs through specific technologies, database schemas, and interface layouts.

2. High-Level Architecture

2.1 Architectural Pattern

SmartStream operates on a **Client-Server Architecture** utilizing the **MVC (Model-View-Controller)** design pattern. Rather than maintaining separate, disparate applications for desktop and mobile users, the system is engineered as a single, highly responsive web application. This approach ensures that updates are universally applied without requiring field staff to install application packages manually. The MVC pattern strictly separates the presentation layer (View) from the business logic (Controller) and data management (Model), promoting code reusability and scalability.

2.2 Technology Stack Justification

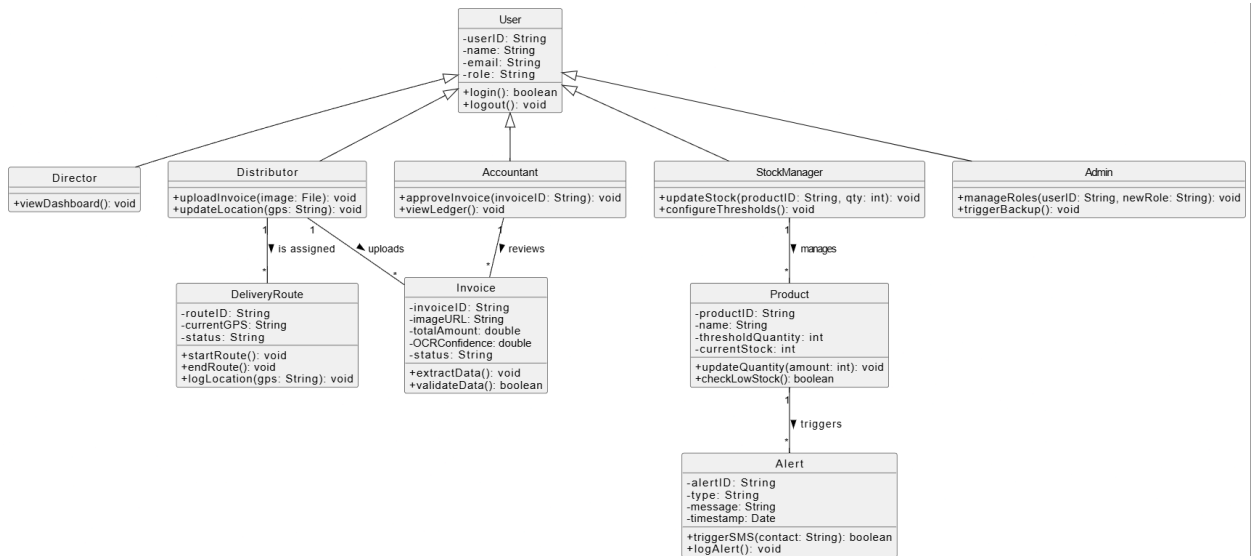
- **Frontend: React with TypeScript:** React's component-based architecture is ideal for rendering dynamic, high-performance user interfaces, such as the Director's live executive

dashboard and the field distributor's mobile forms. TypeScript provides strict static typing, which drastically reduces runtime errors and improves IntelliSense capabilities during collaborative development.

- **Backend: Node.js + Express:** Node.js employs an asynchronous, non-blocking I/O execution model. This makes it exceptionally efficient at handling high volumes of concurrent API requests—a critical requirement for SmartStream, as the server must process incoming GPS coordinate streams from multiple distributors while simultaneously routing heavy AI OCR image processing workloads. Express acts as the lightweight framework to scaffold our RESTful API endpoints.
- **Database: PostgreSQL:** PostgreSQL is selected as the central data store due to its robust ACID compliance and strict relational data integrity. Given that SmartStream processes sensitive financial ledgers, inventory deductions, and delivery verifications, maintaining transactional safety is paramount. It also offers advanced indexing for querying complex analytical dashboard reports rapidly.
- **Storage: AWS S3 (Amazon Simple Storage Service):** AWS S3 provides virtually limitless, secure, and cost-effective object storage. It is integrated specifically to handle the thousands of high-resolution invoice images captured by field staff. By storing these heavy blob assets in S3 and only keeping lightweight reference URLs in PostgreSQL, the primary database remains highly performant.

3. Object-Oriented Design (UML Class Diagram)

The system's structural logic is mapped using a unified UML Class Diagram. A foundational `User` class dictates the core authentication properties (userID, name, email) and enforces Role-Based Access Control (RBAC) through inheritance by specific actor classes (Director, Distributor, StockManager, Accountant, Admin). Core entities such as `Product`, `Invoice`, and `DeliveryRoute` are mapped with specific attributes (e.g., `OCRConfidence` for AI validation) and methods representing their business capabilities.

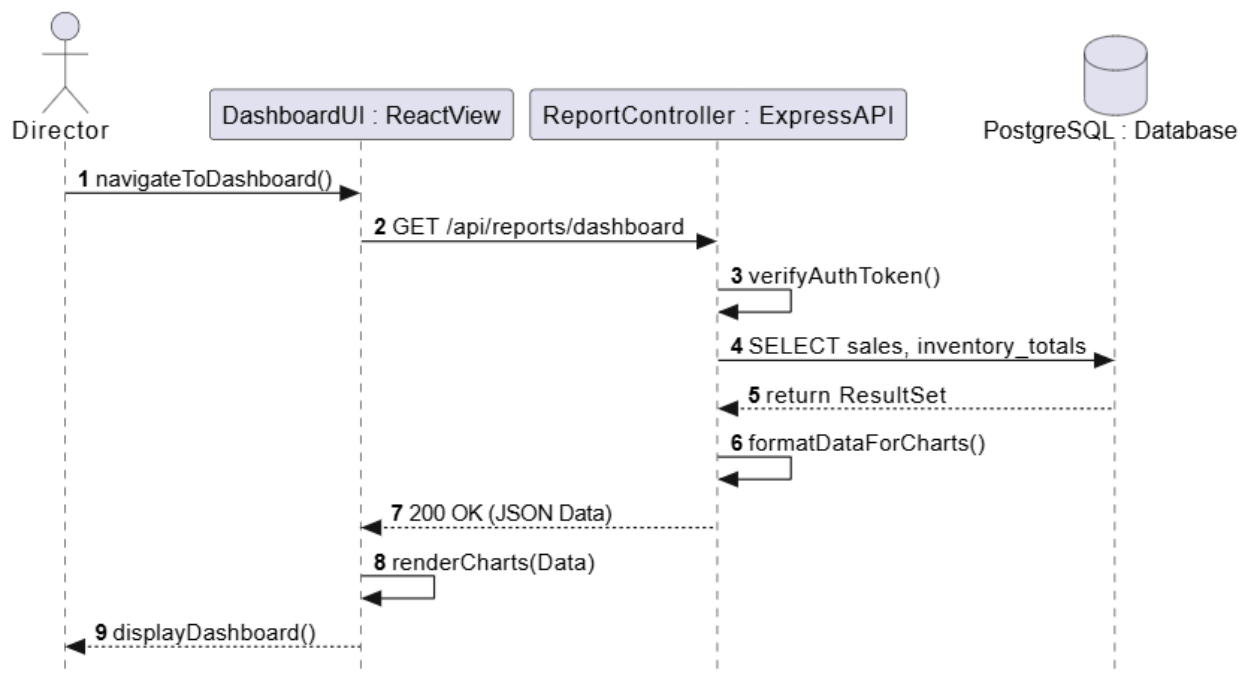


4. Dynamic Behavior & MVC Interaction

The following Design Sequence Diagrams realize the use cases by breaking the "System" into its constituent technical components: The React Views, the Express API Controllers, and the PostgreSQL/AWS S3 Storage models. These diagrams illustrate synchronous API calls, database queries, and asynchronous service interactions.

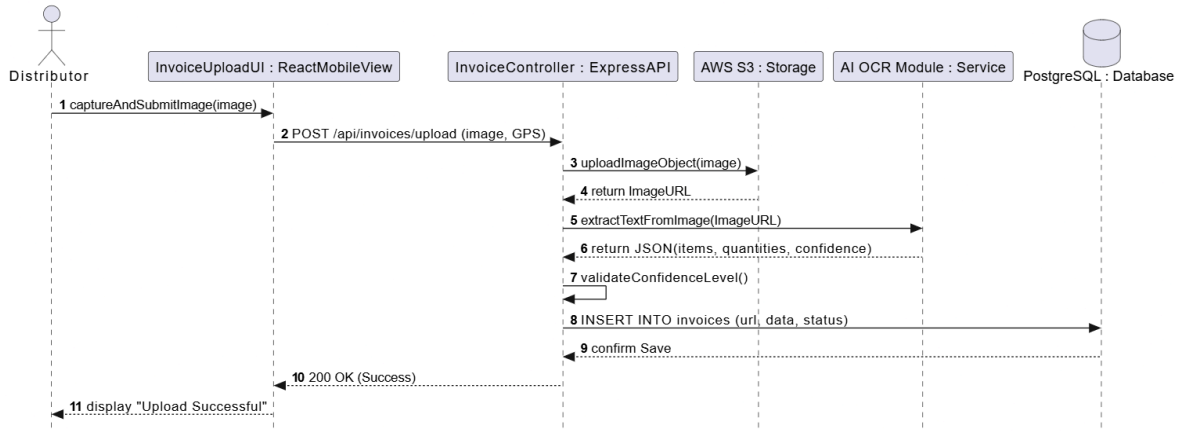
4.1 View Executive Dashboard (Director)

This diagram illustrates the flow where the ReactUI initiates a secure GET request for dashboard metrics. The Express API verifies the authentication token, executes analytical SELECT queries against PostgreSQL, formats the result sets into chart-ready JSON structures, and returns the payload to the frontend.



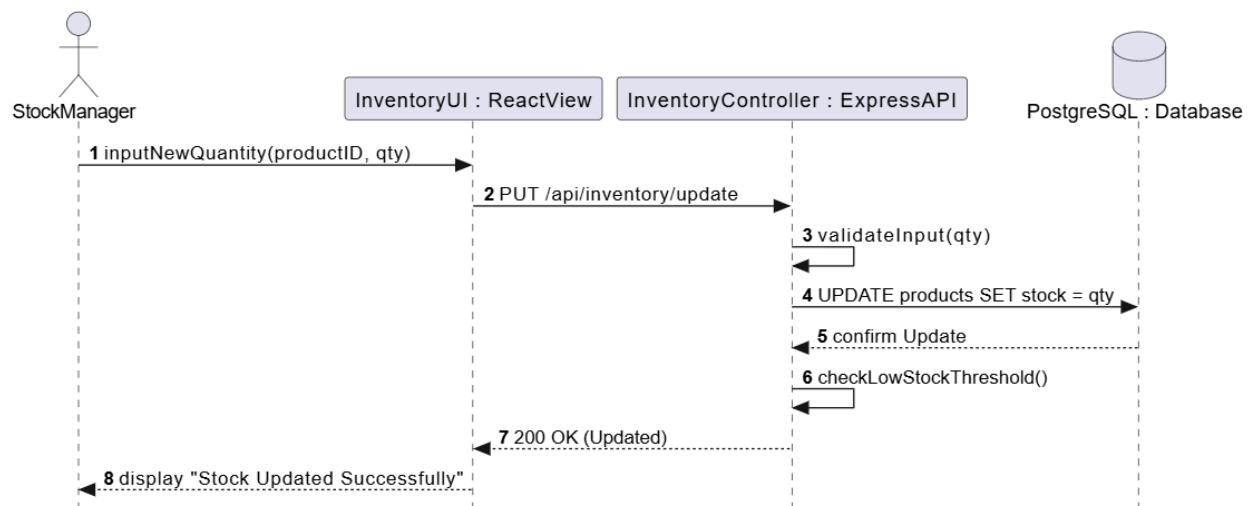
4.2 AI OCR Invoice Upload (Distributor)

This complex flow maps the distributor capturing an image via the mobile React view. The controller uploads the raw binary file to AWS S3, retrieves the object URL, and submits it to the external AI OCR service. Upon receiving the extracted text and confidence scores, the controller validates the data and executes an INSERT operation into the PostgreSQL database.



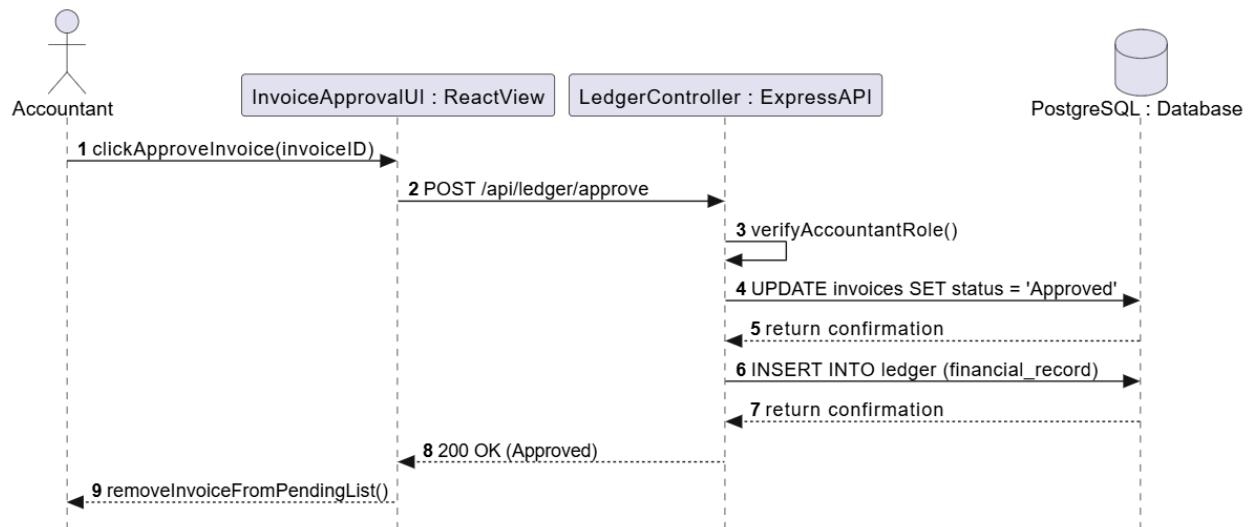
4.3 Manage Inventory Levels (Stock Manager)

Demonstrates the inventory update process via a PUT request. The Node.js controller validates the input quantities, updates the PostgreSQL stock ledger, and independently executes a logical check against the predefined low-stock threshold to trigger an asynchronous alert if necessary.



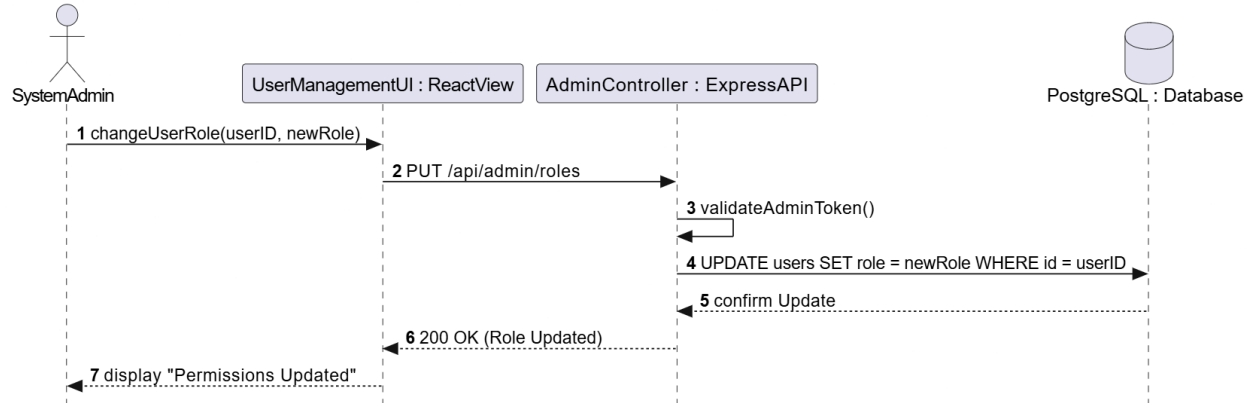
4.4 Approve Processed Invoices (Accountant)

Maps the Accountant's review workflow. The system processes an approval POST request, verifies the user's role-based permissions, and executes sequential database updates to both the `invoices` table (status change) and the `ledger` table (financial record creation).



4.5 Manage User Roles (System Admin)

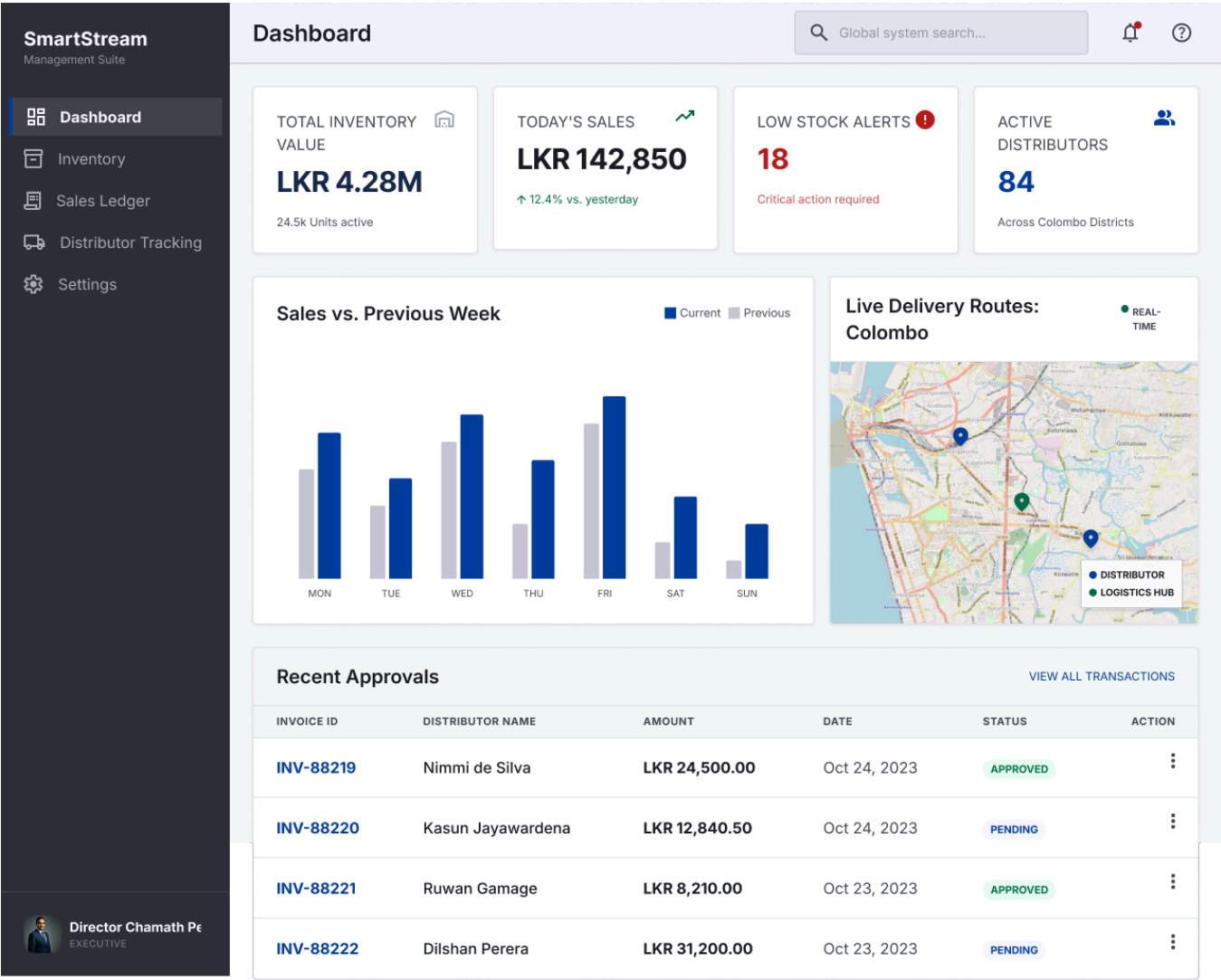
Details the administrative function of managing RBAC. The controller validates a super-admin session token before permitting an UPDATE statement to alter a target user's role capabilities within the PostgreSQL database.



5. User Interface Wireframes

5.1 Desktop Web View: Executive Dashboard

Designed for high-resolution screens utilized by the Director and office staff. It features a left-aligned navigation sidebar and a grid-based main view containing live KPI cards, interactive sales charts, and a real-time GPS tracking map overlay.



Pending Invoice Approvals

QUEUE: 14



Ref: #INV-2024-089 >

RAMEN & ICE CREAM MANIFEST SCAN



Manifest Scan Details	
Item	Quantity
Shin Ramyun Bulk Pack	45
Vanilla Bean Ice Cream 5L Tub	12
Cold Chain Delivery Surcharge	1
Subtotal: LKR 12,915.00	
Tax (8.5%): LKR 1,097.78	
Total Payable: LKR 14,012.78	

Data Verification

Validated via SmartOCR Engine 4.2

CONFIDENCE SCORE **98.4%**

VENDOR ENTITY

Global Ramen Distributors

INVOICE DATE

24 August 2024

EXTRACTED LINE ITEMS

[Add Row](#)

DESCRIPTION	QTY	UNIT PRICE	TOTAL	
Shin Ramyun Bulk Pack	45	LKR 245.00	LKR 11,025.00	✓
Vanilla Bean Ice Cream 5L Tub	12	LKR 120.00	LKR 1,440.00	✓
Cold Chain Delivery Surcharge	1	LKR 450.00	LKR 450.00	✗

* Highlighting discrepancies in Logistics Fee against contract rates.

Subtotal	LKR 12,915.00
Tax (8.5%)	LKR 1,097.78
Total Payable	LKR 14,012.78

Flag for Manual Review

Approve to Ledger

Food & Dessert Inventory

Search inventory...

Export Report



TOTAL ITEMS 1,248	LOW STOCK 12	IN TRANSIT 42	TOTAL VALUE LKR 428k
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PRODUCT ID	ITEM NAME	CURRENT STOCK	THRESHOLD LEVEL	STATUS	ACTIONS
INV-9821	Baskin Robbins Mint Chocolate Chip (1L)	450 units	100	Healthy	
INV-7430	Samyang Buldak Spicy Ramen (5-pack)	12 units	50	Low Stock	Update Stock
INV-1102	Baskin Robbins Pralines 'n Cream (500ml)	89 units	80		
INV-4456	Nongshim Shin Ramyun (Single Cup)	5 units	20		
INV-0092	Baskin Robbins Very Berry Strawberry (1L)	1,024 units	200	Healthy	

Showing 1-5 of 1,248 results

< 1 2 3 ... 250 >

Update Quantity

New Stock Level

12

Save Cancel

Incoming Shipments

	Batch #BR-ICE-01 (Assorted Tubs) Estimated delivery: Tomorrow, 10:00 AM	In Transit
	Ramen Restock - Spicy Series Processing at HUB-A	Pending

Stock Intelligence

Based on current sales velocity, 3 flavors are projected to reach critical levels within 72 hours.

Optimization Score 92%

View Projections

SmartStream

MANAGEMENT SUITE

Dashboard

Inventory

Sales Ledger

Distributor Tracking

Settings

KD

Chamath Perera

System Admin

System Configuration & RBAC

Search system logs...

System Health & Resources

Real-time status of critical infrastructure

DATABASE STATUS

Online

Latency: 14ms | 99.9% Uptime

AWS S3 STORAGE

45% Full

922 GB of 2.0 TB used

API Gateway

Redis Cache

Trigger Manual Backup

GLOBAL ACTIVITY

User Role Modified

Admin updated 'Sara Chen' to Manager

2 mins ago

Backup Success

Automated daily backup completed

4 hours ago

Security Alert

Failed login attempt from IP 192.x.x.1

6 hours ago

View Audit Logs

User Management

124 Total

Invite User

NAME	EMAIL	STATUS	CURRENT ROLE	ACTIONS
Pathum Nissanka ID: ACC-8902	m.thorne@gmail.com	Active	System Admin	
Dilhara Fernando ID: ACC-4115	e.rodriguez@gmail.com	Active	Lead Manager	
Niroshan Dickwella ID: ACC-7721	c.wei@gmail.com	Offline	Operator	
Anusha Kumari ID: ACC-1092	s.jenkins@gmail.com	Active	Viewer	

Showing 1 to 4 of 124 users

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1

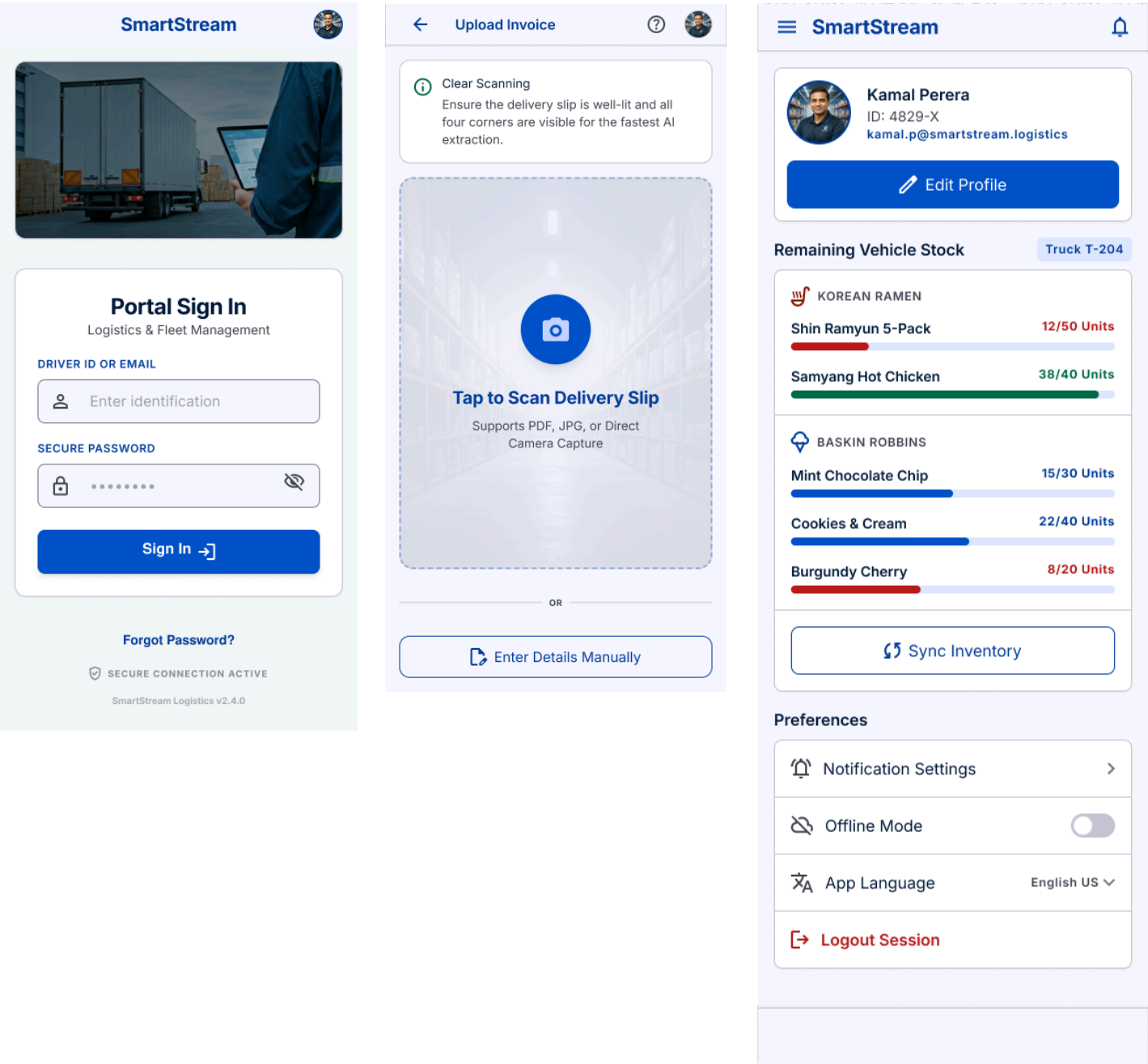
2

3

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5.2 Mobile Responsive Web View: Distributor Upload App

Engineered specifically for field workers operating smartphones single-handedly. The interface maximizes touch target sizes, relies heavily on primary CTA buttons (like the AI Scanner activation), and utilizes high-contrast visual cues to indicate successful data transmissions or active GPS tracking states.



[Wireframe](#)